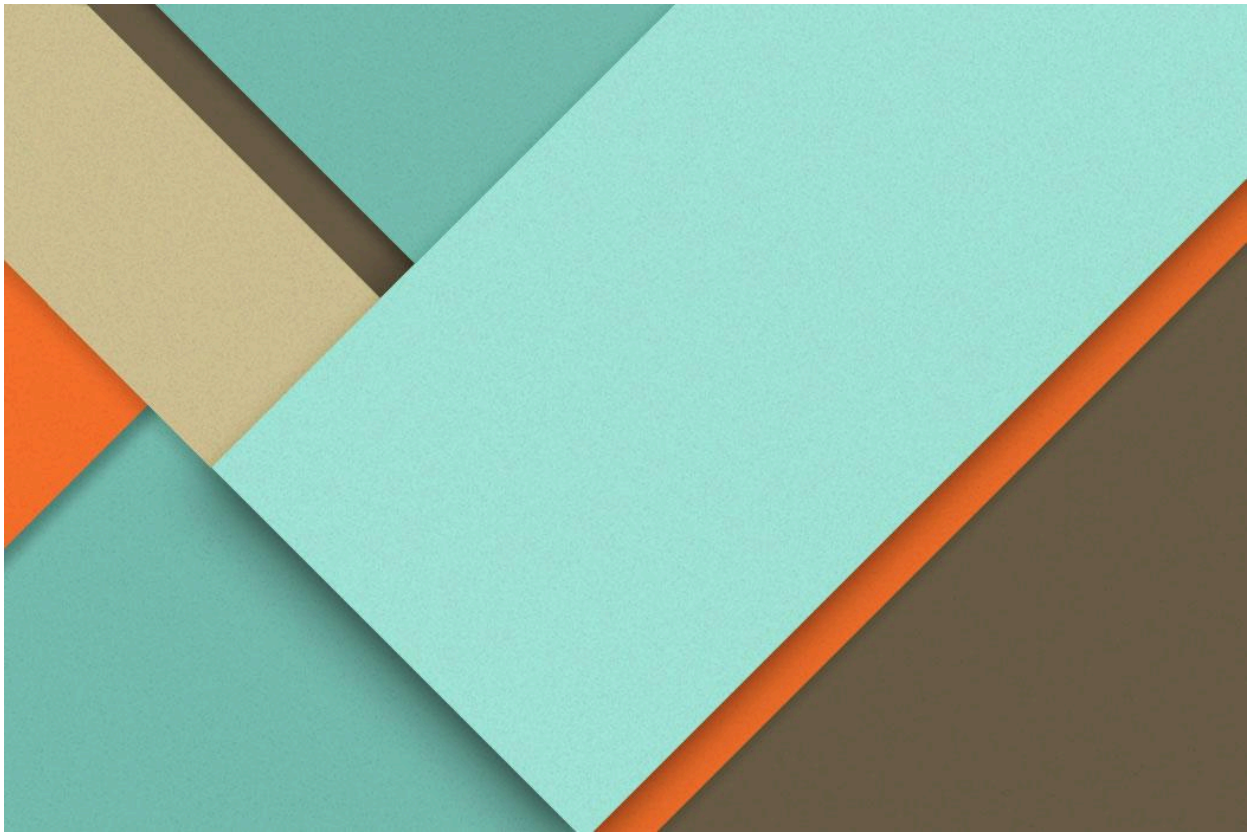

Coach

v 1.0

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Overview

The **Student Knowledge Manager** is an AI-based platform designed to help school and college students manage their study materials, notes, and learning activities in a smart and organized way, Student freely access world wide creative/unique Freature.

Goals

The main goal of the **Student Knowledge Manager** project is to develop an intelligent platform that helps students manage, understand, and organize their study materials more efficiently. The system aims to simplify the learning process by using modern technologies such as **AI and ML.**

Main Goals

1. **Centralized Study Platform**
2. **Smart Content Understanding**
3. **Efficient Study Management**
4. **Time Saving Learning System**
5. **Better Exam Preparation**
6. **User-Friendly Learning Experience**
7. **Improve Learning Productivity**

Why

Most students spend a lot of time searching for important information in lengthy study materials. This process is time-consuming and can reduce learning efficiency. Additionally, many existing platforms only provide basic note-taking or file storage features but do not offer intelligent support for understanding and managing knowledge.

Existing tool's Feature/Service

1 Notes & Knowledge Management	Text notes create karva Rich text editor (bold, table, code block) Subject-wise notes organize Folder / notebook system Notes edit / delete Version history (old version restore) Markdown support
2 File & Document Management	PDF upload DOC / PPT upload Image upload Lecture slides store Large file storage Cloud backup File preview without download
3 Smart Search System	Keyword search Full document search Tag based search Filter by subject / topic AI semantic search
4 Tagging & Organization	Notes tagging (#math #physics) Topic classification

	Custom labels Smart folders
5 AI Features (New Platforms)	AI summary generator AI note generator AI writing assistant AI question answering AI concept explanation AI automatic highlighting AI content recommendation
6 Study Planning System	Study planner Daily study schedule Weekly learning plan Goal setting Study reminders
7 Flashcards & Revision	• Flashcard creation • Memory training • Spaced repetition system • Quick revision mode (aa feature Quizlet ma bahu famous che)
8 Quiz & Test System	• Automatic quiz generation • MCQ test • Practice exams • Instant result

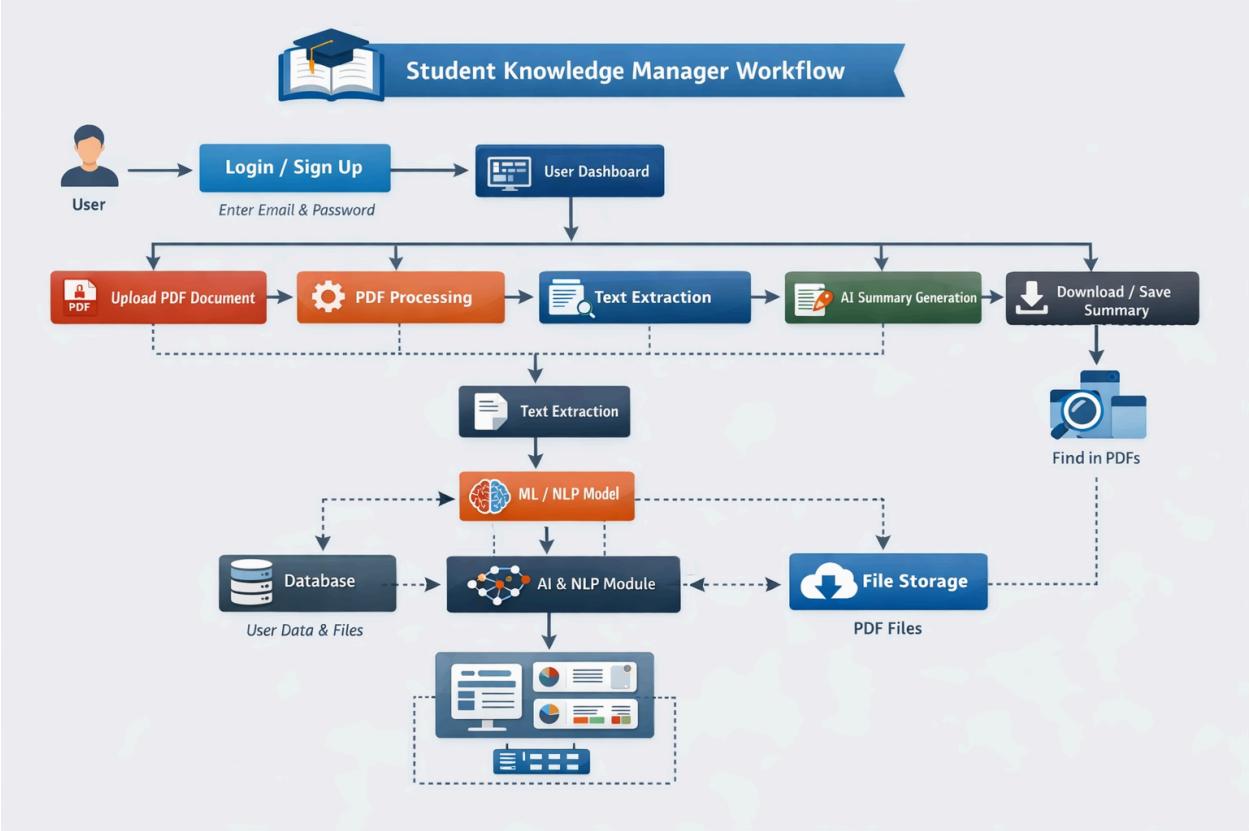
	• Weak topic detection
9️⃣ Collaboration System	• Notes sharing • Group study workspace • Team collaboration • Comment system (aa feature Google Drive ane Microsoft OneNote ma strong che)
🔟 Knowledge Graph System	• Notes ne ek bija sathe connect karvu • Topic relation visualize karvu • Knowledge map create karvu (aa feature Obsidian ma famous che)
1️⃣1️⃣ Web Integration	• Web clipper (website mathi note save) • Bookmark system • Online article save (aa feature Evernote ma available che)
1️⃣2️⃣ Productivity Tools	To-do list Task manager Assignment tracker Deadline reminder
1️⃣3️⃣ Analytics & Progress Tracking	Study time tracking Topic progress tracking

	Learning analytics Performance dashboard
14 Mobile & Cloud Features	Mobile app access Cross-device sync Offline access Auto backup
15 Security Features	Account login Data encryption Private notes Access control
16 Export & Sharing	PDF export Document export Share link Public notes

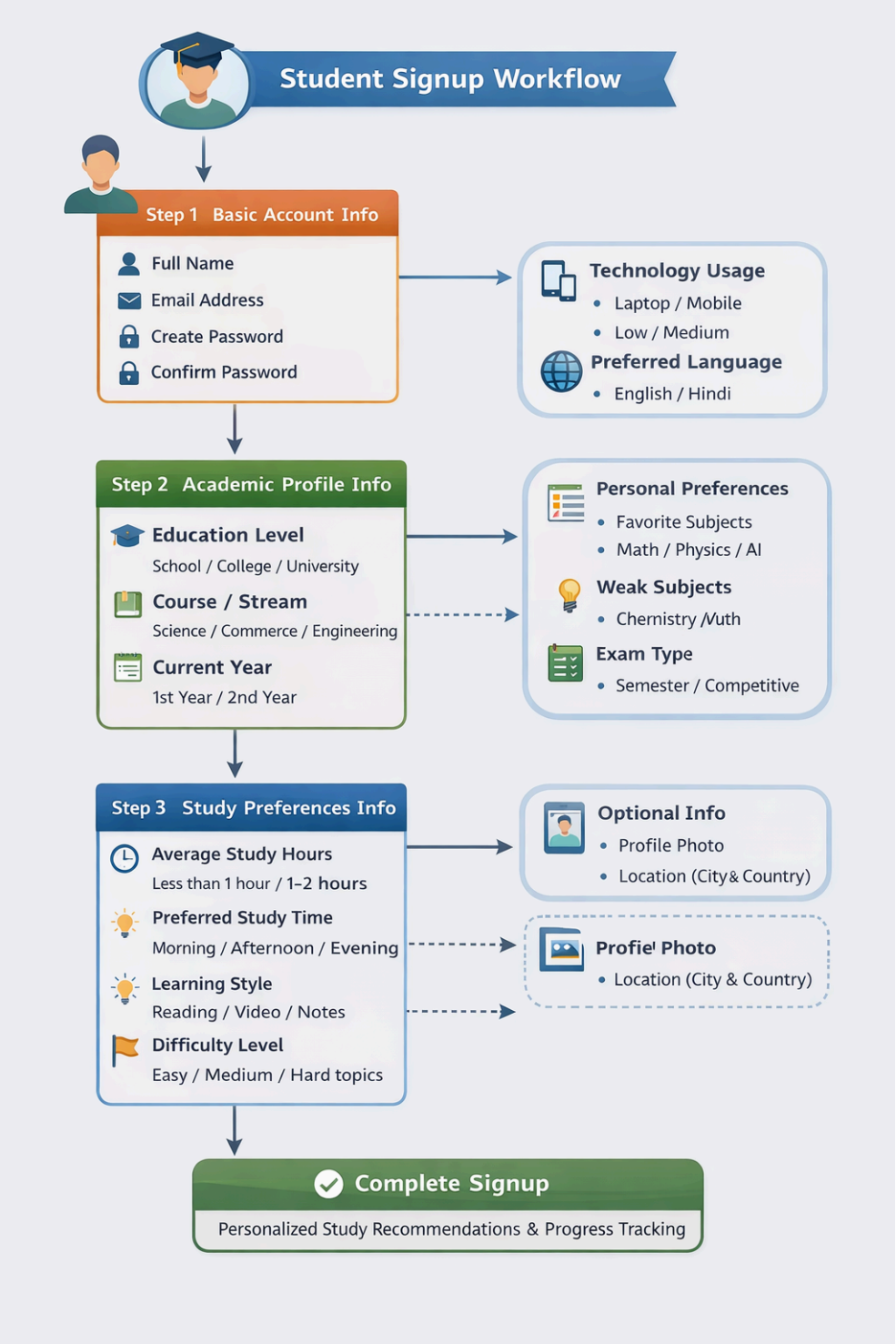
My tool Main Feature/Service

- Upload PDF Document
- Generate Summary for particular pdf
- Search bar (student easily findout word's/topic)

General Workflow/Flowchart



Main Workflow/Flowchart...



Main Workflow/Flowchart...

More info...

1 Project Overview

Tamari website ek **AI based student tool** hase jema students potani study PDF upload kari ne teni **automatic summary generate kari sakse**.

Aa system mainly **Machine Learning** ane **Natural Language Processing** technology use karse.

Purpose:

- Students nu study easy banavvu
- Long notes ne short summary ma convert karvu
- Important topics quickly find karva

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2 Complete Website Modules

1 Landing Page

Landing page par aa elements hase:

Features:

- Website introduction
- Project benefits
- Login / Sign up button

- Feature explanation

Sections:

- Hero section
- About tool
- Features
- How it works
- Contact section

2️⃣ Authentication System

User account banavva mate system.

Features:

- Sign Up
- Login
- Logout
- Forgot password

Database store karse:

- Name
- Email
- Password

- User ID

3️⃣ User Dashboard

Login pachhi user ne dashboard dekhashe.

Dashboard ma:

- Upload PDF
- My Documents
- Generate Summary
- Search Topic
- Profile settings

4️⃣ PDF Upload Module

User PDF upload karse.

Process:

User → Upload file → Server store → Database entry

Possible file limit:

- Max size: 10MB
- Format: PDF only

5️⃣ PDF Text Extraction Module

Uploaded PDF mathi text extract thase.

Tools:

- Python libraries
- OCR (future ma)

Libraries example:

- PyPDF
- PDFMiner

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6 AI Summary Generator

System PDF ne analyze kari ne **short summary generate karse**.

Process:

PDF text → NLP model → summary

Models:

- Transformer models
- Text summarization algorithms

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7 Search Engine

User koi word search kare.

System:

- Uploaded PDF ma search karse

- Related sentence show karse

Example:

Search: "Machine Learning"

Result:

Show paragraph jya aa word che.

8️⃣ Database System

Store karse:

Users table

Field	Description
UserID	unique ID
Name	user name
Email	login email
Password	encrypted

Documents table

Field	Description
DocumentID	file ID
UserID	owner

FileName	PDF name
UploadDate	date

Summary table

Field	Description
SummaryID	summary ID
DocumentID	linked PDF
SummaryText	generated summary

---

3 Technology Stack

Frontend

Languages:

- HTML
- CSS
- JavaScript

Framework:

- React

Purpose:

- User interface

- Dashboard
- Upload UI

Backend

Language:

- Python

Framework:

- Flask
  or
- FastAPI

Machine Learning

Libraries:

- Transformers
- NLTK
- spaCy

Database

Options:

- MongoDB
  or
- MySQL

Cloud Storage

Options:

- Amazon Web Services S3
- Google Cloud storage

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4 Complete Website Pages

Website ma aa pages hase:

1 Home Page
2 Login Page
3 Signup Page
4 User Dashboard
5 Upload PDF Page
6 Summary Result Page
7 Search Result Page
8 Profile Page

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5 System Architecture

```
User Browser
   |
   ▼
Frontend (React)
   |
   ▼
Backend API (Flask / FastAPI)
   |
┌──────────────────────┐
|            |
▼            ▼
Database      AI Model
(MongoDB)   (NLP Summarizer)
   |
   ▼
File Storage
(PDF Documents)
```

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6 Security Features

- Password encryption
- Secure file upload
- File type validation
- Authentication token